

Northwind Traders – Power BI Dashboard Project

Dataset: Northwind Traders

Create an interactive **Power BI Sales & Business Performance Dashboard** using the Northwind Traders dataset.

1. Data Import & Exploration

- Import the Northwind dataset into Power BI.
- Explore the available tables and understand the purpose of each table.
- Identify the relationships between tables.

2. Data Cleaning & Transformation – Power Query

Use **Power Query** to prepare the data.

You should perform appropriate transformations, such as:

- Handle missing/null values.
- Remove unnecessary columns.
- Remove duplicates where appropriate.
- Correct data types.
- Rename columns where necessary.
- Format date and numeric columns.
- Perform **Unpivot** or other transformations if they are useful for your analysis.

3. Data Modeling

Build an appropriate data model for the Northwind dataset.

- Create relationships between the tables.
- Check the relationship cardinality.
- Identify fact and dimension tables where appropriate.
- Use relationships correctly when building your visuals and DAX calculations.

4. Calculated Column

Create at least **one useful Calculated Column** using **DAX**.

For example, you could calculate something related to:

- Product sales
- Profit
- Order value
- Product/category classification

5. DAX Measures

Create at least **3 DAX Measures**.

Your measures should answer useful business questions.

Examples include:

- Total Sales
- Total Orders
- Total Quantity
- Average Order Value
- Total Profit
- Profit Margin

You should demonstrate functions such as:

`SUM()` • `SUMX()` • `AVERAGE()` • `DISTINCTCOUNT()` • `DIVIDE()`

6. Dashboard Visualizations

Create a clear and professional dashboard using **different types of visuals**.

Your dashboard should include:

- **KPI Cards**
- **Column/Bar Chart**
- **Line Chart**
- **Donut/Pie Chart**

- **Table or Matrix**
- At least one additional appropriate visualization

The visuals should help answer questions such as:

- What are the total sales?
- Which products/categories perform best?
- Which customers generate the most sales?
- Which employees handle the most orders?
- How do sales change over time?
- Which countries/regions perform best?

7. Slicers & Filters

Add **interactive slicers** to your dashboard.

Examples:

- Order Date
- Product Category
- Product
- Customer
- Employee
- Country

The slicers should interact with the dashboard visuals.

8. Visual Interactions

Configure interactions between your visuals.

For example:

Select a Category → other charts and KPI cards update accordingly.

Make sure the dashboard allows the user to explore the data interactively.

9. Drill-Through

Create at least **one Drill-Through page**.

Example:

Main Dashboard → Select a Product → Product Details

The drill-through page could display:

- Product information
- Total sales
- Quantity sold
- Number of orders
- Sales trend
- Customers who purchased the product

Add a **Back button** to return to the main dashboard.

10. Business Insights

Based on your analysis, provide at least **3–5 key insights**.

Your insights should be supported by your visuals and calculations.

For example:

Which category generates the highest revenue?

Which products have the strongest performance?

Which customers contribute the most to sales?

What is the overall sales trend?

Which countries have the highest sales?

Final Dashboard Checklist

Your project must contain:

Deliverables

1. Power BI File

- Submit the completed .pbix file.

2. LinkedIn Post

The LinkedIn post should include:

- Short description of the project
- Dataset used: **Northwind Traders**
- Power BI techniques/features used
- 3–5 key insights
- Screenshot of the dashboard
- Mention/tag the relevant person or organization
- Encourage people to check out the project